



Dynamic reliability analysis of main shaft bearings in wind turbines

Zhiyuan Jiang^a, Xianzhen Huang^{a,*}, Huizhen Liu^a, Zhiqun Zheng^a, Shangjie Li^a, Shanshan Du^b

^a School of Mechanical Engineering and Automation, Northeastern University, Shenyang 110819, PR China

^b Wind Power Bearing Business Department of Wafangdian Bearing Group Co., Ltd, Wafangdian 116300, PR China

ARTICLE INFO

Keywords:

Wind turbine
Dynamic reliability
Meta-heuristic algorithm
Tapered roller bearing
Assembly preload

ABSTRACT

Tapered roller bearings (TRBs) are widely employed in large wind turbines as main shaft supports. The reliability of TRBs is directly related to the operational efficiency and safety of wind turbines. In this paper, a method for dynamic reliability analysis of TRBs used for supporting the main shafts of wind turbines is presented. First, a new quasi-static TRB model is established based on the meta-heuristic optimization algorithm. Then, the influence of preload, bearing clearance, and friction on the fatigue life of the TRB is investigated in detail. Moreover, considering the impact of uncertain factors, a dynamic reliability analysis of TRBs under fatigue load is conducted based on the composite limit state and adaptive double-loop Kriging theories. The effect of preload on the dynamic reliability of TRBs is assessed to guide the main shaft bearing assembly of wind turbines. Finally, a practical application example is provided to demonstrate the effectiveness and benefits of the proposed method.

1. Introduction

Conversion from fossil to regenerative energy has become a significant strategy in global green development. Wind power is one of the most potential regenerative energy alternatives [1,2]. The accumulative installed capacity of wind turbines has been increasing rapidly in recent years, especially in off-shore locations where abundant wind resources are available [3,4]. With increased wind turbines and relocation to less accessible off-shore sites, the operation and maintenance (O&M) costs are expected to grow significantly [5]. According to the survey, O&M costs account for 20–30% of the life-cycle cost in land-based wind turbines and 30% in off-shore wind turbines [6]. Therefore, implementing reliability programs for wind turbines to reduce O&M costs is significant.

The reliability of wind turbines can be improved by predicting the weak points in the system through statistical data of failures and strengthening them at the design stage. Ossai et al. [7] quantified the component maintenance downtime as a percentage of life cycle duration, and developed a maintenance strategy for wind turbines based on the six-stage Markov model. Kang et al. [8] investigated the failure rate and the mean time to failure of wind turbine systems. Moreover, the authors established a reliability model of wind turbines based on the Fault Tree Analysis (FTA). Márquez et al. [9] evaluated the reliability of wind turbines combined with the Binary Decision Diagram (BDD)

method and FTA. Zhang et al. [10] implemented the Fuzzy Fault Tree Analysis (FFTA) to assess wind turbine failure intervals and component importance. Bhardwaj et al. [11] investigated the reliability of off-shore wind turbine gearboxes based on Failure Mode and Effects Analysis (FMEA). In addition, the authors determined the critical components that cause malfunctions of wind turbines. Li et al. [12] presented a Bayesian Networks (BNs) model to assess the reliability of wind turbines. The authors reported that the BNs model is more precise than the FTA. Sinha et al. [13] developed a Failure Modes Effects and Criticality Analysis (FMECA) to evaluate the failure of the gearbox in an offshore wind turbine. Scheu et al. [14] evaluated the failure of offshore wind turbines installed around Europe based on the FMECA. Li et al. [15] presented an improved FMEA method for the support structures of wind turbines. The authors proved that the proposed method eliminates the defects of conventional FMEA. The methods mentioned above can be used to analyze potential hazards of wind turbines, evaluate failure models of critical systems and components, rank prioritization of the failure model, and further provide corrective or preventive recommendations for reliability allocation and O&M management [16,17]. Nevertheless, it is relatively difficult to obtain sufficient failure data for developing reliability models in real-world applications [18,19]. Moreover, within the existing investigations, the failure mechanisms of structural components of wind turbines are not adequately explored. Hence, they can hardly provide essential guidance for designing,

* Corresponding author.

E-mail address: xzhhuang@mail.neu.edu.cn (X. Huang).

<https://doi.org/10.1016/j.ijmecsci.2022.107721>

Received 27 May 2022; Received in revised form 28 August 2022; Accepted 5 September 2022

Available online 8 September 2022

0020-7403/© 2022 Elsevier Ltd. All rights reserved.

논문 포인트

1. ISO 16281 Basic reference rating life 기준 수명 평가
 - 윤활, 오염도 내용은 복잡하다고 생각
 - 대신 Preload, clearance 등의 영향 on 수명 중점 확인
2. Dynamic reliability analysis를 통해 시간에 따른 파손 확률을 모델링하고, 파라미터 스터디 수행
 - 특별한 필요성, 이점은 잘 모르겠음 (논문 추가 리딩 필요)

manufacturing, and assembly of wind turbine components.

Many investigations have focused on the physics-model-based method to investigate the failure mechanism of critical components and provide references for wind turbine design. Kim et al. [20] developed a reliability analysis framework for jacket-type offshore wind turbine support structures based on the response surface method (RSM) and the first order reliability method (FORM). Wandji et al. [21] developed the reliability-based design method for semi-floater sub-structures of offshore wind turbines. Yang et al. [22] proposed a reliability-based design optimization (RBDO) method for offshore wind turbine tripod sub-structures. The authors demonstrated that the proposed method could obtain a reliable design with excellent structural performance. Velarde et al. [23] established a fatigue damage model of a large monopile supporting the wind turbine. Furthermore, the authors assessed the fatigue failure probability of a large monopile for a kind of 10 MW wind turbine. Fu et al. [24] presented the probability density evolution method to investigate the fatigue failure probability of wind turbine bolt and flange subjected to random wind load. Lee et al. [25] proposed the RBDO method for a monopile transition piece in an offshore wind turbine to obtain a reliable grouted connection under a short design cycle. Thapa et al. [26] presented a method for uncertainty quantification and global sensitivity analysis of composite wind turbine blades based on polynomial chaos expansion. Toft et al. [27] established a framework to evaluate the reliability of wind turbine blades. Moreover, the authors investigated the effect of blade defects on wind turbine reliability. Liu et al. [28] explored the reliability of blades in a wind turbine with floating foundation support. In addition, the authors also analyzed the effect of support types and material on blade reliability. Su et al. [29] developed an experimental method to assess the reliability of composite wind turbine blades considering material aging. In addition, the authors optimized the structure to improve the reliability of the blades. Liu et al. [30] presented an RBDO method for a wind turbine gear transmission from the perspective of dynamic fatigue reliability sensitivity to design a reliable structure with minimum volume. Nejad et al. [31] derived a method for determining a gear's bending fatigue damage probability. Yan [32] presented a detailed fatigue reliability method for the gear transmission system of a 5 MW wind turbine subjected to a random load process. Dong et al. [33] analyzed the fatigue reliability of gear transmission systems considering surface and sub-surface pitting. The leading reliability research based on the physics-model-based method is mainly focused on the supporting structures, connecting parts, blades, and gearboxes. Nevertheless, applications of reliability analysis on bearings such as pitch, yaw, and main shaft bearing are rarely reported.

The existing studies regarding bearings of wind turbines are concentrated on deterministic analysis and fault diagnosis. He et al. [34, 35] established a new yaw bearing model based on the finite element method. Furthermore, the authors investigated the effect of hardened depths on contact stress and fatigue life. Wang et al. [36] established static models for different types of pitch bearing considering the clearance. The authors also investigated the effects of geometry and load on the static load-carrying capacity. Schwack et al. [37] tested anti-wear properties for pitch bearing under various lubrication and loading conditions. According to the experimental results, the authors concluded that greases with low base oil viscosities and high bleeding rates show satisfactory wear protection under low oscillating amplitudes. Xu et al. [38] evaluated the fatigue life of yaw bearing with various support foundations. The authors reported that foundations types in wind turbines directly affect the dynamic load of yaw bearing. Yucesan et al. [39] proposed a hybrid physics-informed neural network method for the main shaft bearings to evaluate fatigue damage accumulation. Liu et al. [40] collected vibration signals of pitch bearings to diagnose their fault combined with the empirical wavelet thresholding method and the envelope analysis. Pan et al. [41] developed a method for slewing bearing to extract incipient weak fault features based on circular domain analysis and piecewise aggregate approximation. The accuracy of the results

and rationality of the method was experimentally validated. Guo et al. [42] presented a method for bearing fault detection in the wind turbine by data-driven multiscale dictionary construction. Regardless of the existing studies, the effect of uncertainties on failure has not been fully revealed yet. Furthermore, more than 50% of the O&M costs are related to the premature failure of slewing bearings in the wind turbine [43]. Thus, further investigations on the reliability of bearings are required.

In this paper, critical fatigue life is adopted as a quantitative assessment index of the anti-fatigue performance of wind turbine tapered roller bearing (TRB). Moreover, to remove the empirical initial value selection and low efficiency of the traditional models, a new quasi-static model is established based on the meta-heuristic algorithm. The remainder of this paper is organized as follows. In Section 2, a new wind turbine TRB model is derived based on the slicing technique and vector method. The nonlinear equations are transformed into the optimization of L_2 -norm, while the meta-heuristic optimization algorithm method, Harris hawks optimization (HHO), is used to intelligently search optimization objectives without initial empirical value. In Section 3, the equivalent fatigue damage of TRB is evaluated via ISO standard and the linear Palmgren-Miner rule. Meanwhile, the time-variant performance function of fatigue failure caused by design parameters, manufacturing error, and assembly quality is presented. The dynamic reliability analysis of TRBs under the fatigue duty cycle is established based on the composite limit state (CLS) theory and adaptive double-loop Kriging (ADLK) theory to quantitatively assess the effect of randomness on cumulative fatigue failure under the time-variant mechanism. Numerical results of internal contact load distribution under fatigue loads are provided in Section 4. Furthermore, the efficiency and accuracy of the proposed model are verified by the simulation results of the finite element model, Romax software, and Harris model. The coupling effects between the assembly preload, bearing clearance, and friction on the fatigue life of wind turbine TRB are discussed. Moreover, dynamic reliability assessment results are verified, and the effect of assembly preload on dynamic reliability is investigated. Finally, the main conclusions of this paper are summarized in Section 5.

2. Mechanical model of the TRB

As shown in Fig. 1, the rotor of a modern gearless wind turbine is directly connected to the rotor hub, and the main shaft is supported by two tapered roller bearings (TRBs). As core components of the spindle drive system, the TRBs are subjected to heavy loads and complex operating conditions [44]. The internal contact characteristic of TRBs directly affects the reliability of the wind turbine. In this section, a new mechanical model is established for contact characteristics analysis of TRBs.

2.1. Analysis of contact characteristics

The cross-section of the TRB is shown in Fig. 2, where D is the outside diameter of TRB, d is the bore diameter of TRB, d_m is the bearing pitch

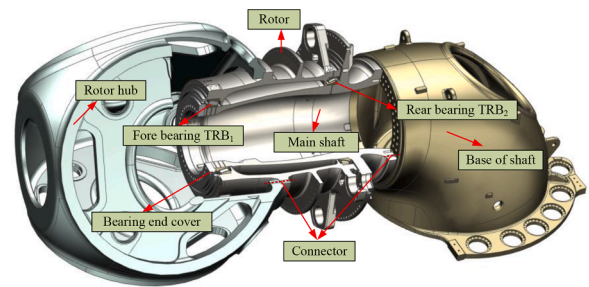


Fig. 1. Schematic diagram of the 5 MW offshore direct drive wind turbine, consisting of the rotor, rotor hub, bearing end cover, main shaft, the base of the shaft, connector, and TRBs.

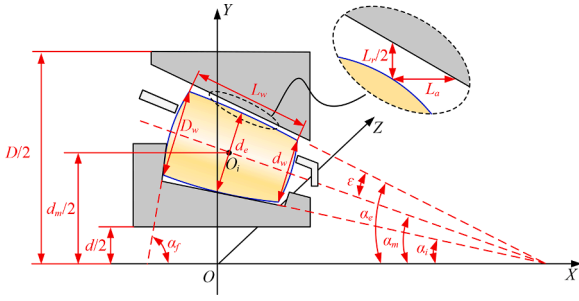


Fig. 2. Schematic diagram of the TRB geometry considering clearance.

diameter, α_e is the contact angle of the outer raceway, α_i is the contact angle of the inner raceway, α_m is the mean contact angle, α_f is the flange angle, ε is the Semi taper angle of the roller, L_w is the effective contact length of the roller, D_w is the roller large-end diameter, d_e is the mean roller diameter, d_w is the roller small-end diameter, O_i is the roller center point, O is the TRB center point, L_r is the radial clearance of the TRB and L_a is the axial clearance of the TRB.

To investigate the contact characteristics of TRB, globe coordinate frames XYZ and XY_rZ_r as well as local cylindrical coordinate frames $U_{xj}U_{yj}\theta$ and $X_{gj}Y_{gj}\psi$, are established, according to Fig. 3. δ_x and δ_{rj} are translational motions, φ_y and φ_z are rotational motions, F_x , F_y , F_z , M_y , and M_z are external forces and moments, and φ_j is the azimuth angle of the j^{th} roller. Specifically, the roller with an azimuth angle of 0 is φ_1 , while the number j increases clockwise.

Fig. 3 illustrates the conversion of the coordinate systems between XYZ , XY_rZ_r , $U_{xj}U_{yj}\theta$, and $X_{gj}Y_{gj}\psi$. Parameters $\delta_{U_{yj}}$ and $\delta_{U_{xj}}$ are displacements along axes U_{yj} and U_{xj} , $\delta_{Y_{gj}}$ and $\delta_{X_{gj}}$ are displacements along axes Y_{gj} and X_{gj} , φ_θ and φ_ψ are misalignment angles in plane $U_{xj}U_{yj}$ and $X_{gj}Y_{gj}$. Parameter φ_{jr} represents the converted angel. Descriptions of the displacements mentioned above are expressed in Appendix A (Eqs. (A.1) - (A.4)).

The normal displacement δ_{ej} of the outer raceway and roller shown in Fig. 4 can be obtained as:

$$\delta_{ej} = \delta_{Y_{gj}}\cos(\varepsilon) + \delta_{X_{gj}}\sin(\varepsilon) \quad (1)$$

where ε is the Semi taper angle of the roller.

The contact area along a roller can be divided into N_s identical slices. The profile function of rollers can be expressed as [45,46]:

$$h(l_k) = K_x d_e \ln\left(1 / \left(1 - (2l_k/L_w)^2\right)\right) \quad (2)$$

where K_x is the coefficient of the crowned profile, d_e is the mean roller diameter, l_k is the position of the k^{th} slice, l_1 is located at the larger roller

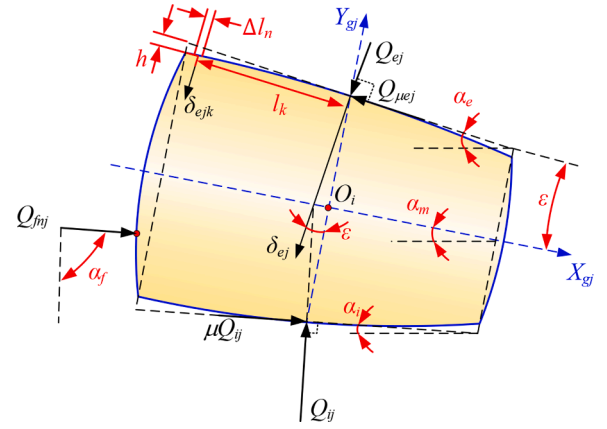


Fig. 4. Schematic diagram of contact characteristics between raceways and the roller, which consists of normal deformation, contact load, and roller profile.

end, and L_w is the effective contact length of the roller. Considering the crowned profile and misalignment angle φ_ψ , the contact deformation of the k^{th} slice δ_{ejk} can be derived as follows:

$$\delta_{ejk} = \delta_{ej} - h(l_k) + (-l_k)\varphi_\psi \quad (3)$$

As shown in Fig. 4, the total contact load Q_{ej} of the roller is the sum of the individual contact load q_{ejk} on each slice and can be expressed as:

$$Q_{ej} = \sum_{k=1}^{N_s} q_{ejk} \begin{cases} q_{ejk} = K_{jk} \delta_{ejk}^{10/9} & \delta_{ejk} > 0 \\ q_{ejk} = 0 & \delta_{ejk} \leq 0 \end{cases} \quad (4)$$

where N_s is the total number of slices on a single roller, and K_{jk} is the stiffness coefficient related to the material and shape of the roller and raceway. Parameter K_{jk} can be calculated according to [47].

Since direct-drive wind turbines operate under heavy loads and low-speed conditions, the fluid-film friction between the roller and the raceway can be neglected [44]. However, the thrust contact friction between the roller and the outer raceway significantly affects the internal load distribution [48]. Therefore, the proposed model considers the frictional force at the roller-raceway contact, as shown in Fig. 5(b). The thrust contact friction between the roller and the outer raceway can be expressed by:

$$Q_{\mu ej} = \sum_{k=1}^{N_s} \mu q_{ejk} \quad (5)$$

where μ is the frictional coefficient. As shown in Fig. 5(c), the friction torque produced by the thrust contact friction can be expressed as:

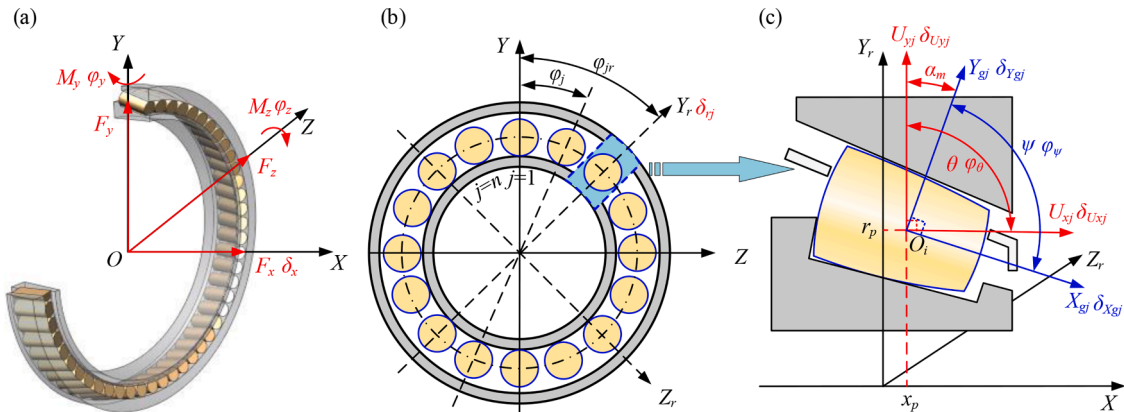


Fig. 3. Schematic diagram of various coordinate frames defined to describe the displacement of the TRB assembly: (a) globe coordinate frames XYZ . (b) converted global coordinate frames XY_rZ_r in radial cross-sectional view, and (c) local cylindrical coordinate frames $U_{xj}U_{yj}\theta$ and $X_{gj}Y_{gj}\psi$ in axial cross-sectional view.

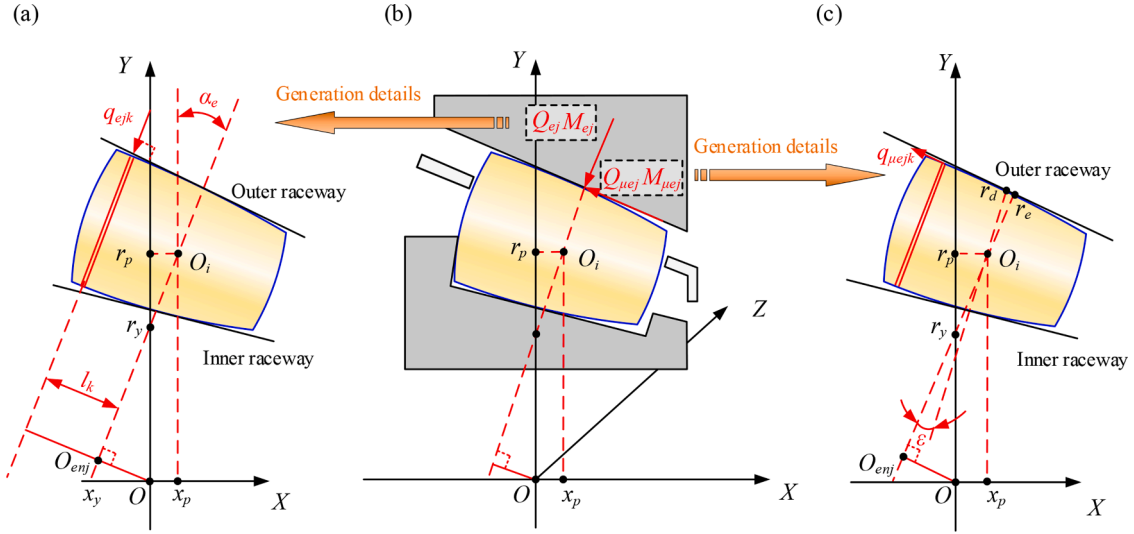


Fig. 5. Interaction between the j^{th} roller and the outer raceway: (a) schematic diagram of the contact load and the corresponding moment for the k^{th} slice, (b) contact characteristics between the j^{th} roller and the outer raceway, and (c) schematic diagram of thrust contact friction and corresponding moment for the k^{th} slice.

$$M_{\mu ej} = \sum_{k=1}^{N_s} q_{\mu ejk} l_{eenj} \quad (6)$$

where $leenj$ is the length from point r_e to O_{enj} , expressed in Appendix A (Eqs. (A.5), (A.6)), and $q_{\mu ejk}$ is the thrust contact friction of each slice.

As shown in Fig. 5(a), the resistance torque produced by the Q_{ej} can be expressed as:

$$M_{ej} = \sum_{k=1}^{N_s} \{ q_{ejk} [(r_p - x_p / \tan(\alpha_e)) \sin(\alpha_e) + (-l_k)] \} \quad (7)$$

To summarize, force balance equations $\mathbf{E}_q$ of the TRB are expressed in Appendix B (Eq. (B.1)). Furthermore, the contact characteristic between the raceway and roller in the local cylindrical coordinate frame $X_{gj}Y_{gj}Z_{gj}$ is shown in Fig. 4. Equations of equilibrium for rollers and raceway are expressed in Eq. (B.2).

2.2. Solution of force balance equations

Equations $\mathbf{E}_q$ described in Eq. (B.1) are highly nonlinear. Gradient-based iterative approaches, such as the Newton-Raphson method, were recommended to solve them [47,49–52]. However, if the initial values of unknown variables are not properly chosen, iterative methods will be difficult to converge [53]. Moreover, gradient-based approaches may encounter serious speed deficiencies for high-dimensional problems, negatively affecting efficiency. A new method is presented in this paper to overcome these problems and solve the highly nonlinear equations from the perspective of meta-heuristic optimization.

In terms of the Lyapunov theorem, the nonlinear systems can be

stabilized as the norm of the state variables converges to zero [54]. Therefore, equations $\mathbf{E}_q$ can be converted to the objective function of an optimization:

$$\text{minimize } \|\mathbf{E}_q\|_2 = \left(\sum_{i=1}^m |E_{qi}|^2 \right)^{1/2} \quad (8)$$

where $\mathbf{E}_q = [E_{q1}, E_{q2}, E_{q3}, \dots, E_{qm}]$ denotes an m -order nonlinear system, as shown in Eq. (8). Moreover, to meet the constraints of geometry and working conditions, unknown variables should be limited to certain ranges, as shown in Table 1. $F_{ra} = \sqrt{F_y^2 + F_z^2} \tan(\alpha_m)$ is the derived axial force of the TRB, c_1 and c_2 are the corner radius of the roller large-end and roller small-end.

Once the objective function and design variable bounds are derived, any optimization algorithm can be employed to search the design variables. In this paper, the formulated problem is solved by the meta-heuristic algorithm called Harris hawks optimization (HHO) [55], which is inspired by the behavior of the Harris hawk in hunting fleeing prey. The HHO algorithm mainly consists of two steps: exploration and exploitation. First, the number of possible solutions is randomly created in the search space. Simultaneously, the escape energy of the prey is calculated. According to the escape energy, the HHO algorithm can switch from exploration to exploitation. In these phases, the Harris hawks choose different chasing strategies based on the escaping behavior of the prey. Finally, the position of the prey is determined when the escape energy is equal to 0. According to the core concepts of the HHO algorithm, design variables δ_x , δ_{rj} , φ_y , and φ_z can be obtained without considering the initial values and gradient information.

Table 1
The calculation formula of basic design variables.

Design variables	Description
δ_x	$\begin{cases} \delta_x = 0 & F_x \leq F_{ra} \\ 0 < \delta_x \leq \sqrt[10]{(F_x - F_{ra})/K_{jk}} & F_x > F_{ra} \end{cases}$
δ_{rj}	$ \delta_{rj} \leq \sqrt[10]{\sqrt{F_y^2 + F_z^2}/K_{jk}}$
φ_y	$-\frac{180K_x d_e \ln\left(1/\left(1 - \left(1 - \frac{2c_2}{L_w}\right)^2\right)\right)}{(L_w/2 - c_2)\pi} \leq \varphi_y \leq \frac{180K_x d_e \ln\left(1/\left(1 - \left(1 - \frac{2c_1}{L_w}\right)^2\right)\right)}{(L_w/2 - c_1)\pi}$
φ_z	$-\frac{180K_x d_e \ln\left(1/\left(1 - \left(1 - \frac{2c_2}{L_w}\right)^2\right)\right)}{(L_w/2 - c_2)\pi} \leq \varphi_z \leq \frac{180K_x d_e \ln\left(1/\left(1 - \left(1 - \frac{2c_1}{L_w}\right)^2\right)\right)}{(L_w/2 - c_1)\pi}$

3. Dynamic reliability of TRB

In previous studies, reliability analysis of bearings is mainly concentrated on axial cracking of raceway [56], which is the domain failure model of high-speed shaft bearings in conventional geared wind turbines [57]. However, as shown in Fig. 1, the generator rotators of gearless wind turbines are directly connected to the turbine rotor hub. The size and weight of the generator are generally taken as large values [58] to satisfy the requirement of power generation. Therefore, the massive TRB carrying the generator is prone to fatigue damage, which is the dominant failure mode of gearless wind turbines [59].

3.1. Fatigue life evaluation

In practical scenarios, the fatigue life of a roller bearing under a determined load case can be estimated based on ISO 16281 or 281 standards. Notably, the fatigue assessment method from ISO 281 neglects the effect of bearing clearance and misalignment [46], which inevitably occur during the assembly process of wind turbines [60]. Hence, in this paper, the service life of the main shaft bearing is assessed according to ISO 16281. Furthermore, due to the long-term exposure to ever-changing weather stages, the internal temperature and contamination status of the wind turbine varies over time [61]. Since these factors are relatively complex to describe, the fatigue model of TRB in this section adopts the basic rating life.

The basic rating life of TRB under a determined load case i can be expressed as:

$$L_{10ri} = \left(\left(\left(\frac{Q_{ci}}{Q_{ei}} \right)^4 \right)^{-1.11} + \left(\left(\frac{Q_{co}}{Q_{eo}} \right)^4 \right)^{-1.11} \right)^{-0.9} \times 10^6 \quad (9)$$

where Q_{ci} and Q_{co} are basic dynamic load ratings of the inner and outer raceway, Q_{ei} and Q_{eo} are the equivalent dynamic loads of a nonrotating and rotating raceway. These parameters are calculated in Appendix C (Eqs. (C.2)-(C.4)). It should be noted that the generator of the direct-drive wind turbine investigated in this paper rotates at low speed since it is directly connected to the turbine rotor hub. Therefore, the outer raceway of the TRB is the rotating raceway.

The bearing on the main shaft experiences a time-varying load, generally compiled to fatigue duty cycle [62]. The fatigue life of TRB under the fatigue duty cycle can be derived in terms of the linear Palmgren-Miner rule [63]:

$$\sum_{i=1}^m \frac{n_i}{N_i} = \sum_{i=1}^m \frac{R_{ii} L_d}{L_{10ri}} = 1 \quad (10)$$

where n_i is the number of cycles experienced under the corresponding load case, N_i is the fatigue life at the corresponding load case, m is the number of load cases in the fatigue duty cycle, L_d is the number of TRB revolutions at the upper fatigue limit with the fatigue duty cycle, R_{ii} is the time ratio of the corresponding load case in the fatigue duty cycle, L_{10ri} is the basic rating life under a determined load case i . It should be noted that R_{ii} equals one when the fatigue duty cycle is composed of a constant load condition.

3.2. Dynamic reliability analysis

Irreversible TRB fatigue degradation will emerge during the service period of wind turbines. The accumulation of the deterioration process may significantly decrease the durability of TRB. From engineering design perspective, the fatigue failure of structure can be considerably altered by multiple uncertainty sources, such as the geometry dimensions, material properties, and load variations [30,64]. Consequently, quantification and propagation of these uncertainties are crucial for the reliability assessment of wind turbine TRB. Moreover, considering cumulative effects, the fatigue failure of the TRB during

wind turbine operation is time-variant. The traditional static reliability approaches without considering the time factor are inappropriate [65–67]. Thus, the dynamic reliability analysis of the wind turbine TRB is presented in this section.

To quantitatively assess the effect of randomness on cumulative fatigue failure, the dynamic reliability of wind turbine TRB can be described as:

$$P_f(0, t_s) = \text{Prob}\{G(\mathbf{X}, t) < 0, \exists t \in [0, t_s]\} \quad (11)$$

where

$$G(\mathbf{X}, t) = L_d(\mathbf{X}) - n \times t \times d_y \times h_d \times m_h \quad (12)$$

where n is the rotational speed of the wind turbine, $\mathbf{X} = [L_r, D, d, K_x, D_w, d_w, L_w, n, \mu]^T$ is the characteristic parameter vector used to estimate the fatigue failure of TRB, t_s is the designed system lifetime of interest. $d_y = 365$, $h_d = 24$, and $m_h = 60$ are constants of days per year, hours per day, and minutes per hour, respectively. Therefore, for time-variant performance function $G(\mathbf{X}, t)$, the domain of failure and the domain of safety can be defined as:

$$\begin{aligned} F &= \{G(\mathbf{X}, t) \leq 0, \exists t \in [0, t_s]\} \\ S &= \{G(\mathbf{X}, t) > 0, \forall t \in [0, t_s]\} \end{aligned} \quad (13)$$

where F is the failure domain, which represents that the fatigue life of the TRB is lower than the design limit, S is the safety domain, which represents the fatigue life of the TRB being in the allowable range. Generally, Eq. (11) cannot be solved by analytical and direct numerical methods due to the high dimensionality and integration complexity.

The composite limit state (CLS) is one of the most accurate methods to dispose of Eq. (11). This method adopts dissection of observation region and assessment of multiple static reliability to define the dynamic reliability [68]. The CLS method for dynamic reliability analysis of wind turbines TRB follows the basic steps:

Step 1: The observation region $T = [0, t_s]$ is discretized to N_t^* time points. The random sample $\mathbf{X}_i = \{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \dots, \mathbf{X}_{N_t^*}\}$ is generated in the probabilistic space based on the joint probability density function $f_{\mathbf{X}}(\mathbf{X})$ of random variable $\mathbf{X}$, which requires the extraction of large-scale samples [69] (generally taken as 10^5) to ensure the calculation accuracy.

Step 2: For the sample point $\mathbf{X}^*$ in the $\mathbf{X}_i$, the fatigue failure of TRB in the observation region $[0, t_s]$ can be expressed as:

$$P_f(\mathbf{X}^*, T) = \text{Prob}\left\{\bigcup_{s=0}^{N_t^*} G(\mathbf{X}^*, t_s) < 0\right\} \quad (14)$$

Step 3: Define the indicator function $I_F(\mathbf{X}_i, T)$ as follows:

$$\begin{cases} I_F(\mathbf{X}^*, T) = 1, & P_f(\mathbf{X}^*, T) > 0 \\ I_F(\mathbf{X}^*, T) = 0, & P_f(\mathbf{X}^*, T) = 0 \end{cases} \quad (15)$$

Step 4: Repeat steps 2 and 3 to assess the $I_F(\mathbf{X}_i, T)$ in the probabilistic space. The cumulative probability of TRB failure can be obtained by:

$$P_f(0, t_s) = \text{Prob}\left(\bigcup_{s=0}^{N_t^*} G(\mathbf{X}, t_s) < 0\right) = \frac{\sum_{i=1}^{N_t^*} I_F(\mathbf{X}_i, T)}{N} \quad (16)$$

3.3. Dynamic reliability analysis based on ADLK

In the CLS method, numerous samples should be generated to cover the observation region and evaluate all instantaneous performance for each design point to ensure the accuracy of TRB dynamic reliability analysis. This process is generally limited by high computing resources requirements [70,71]. Therefore, CLS is often used to verify the probability analysis results of alternative methods [72]. In this section, a new method for probabilistic analysis of TRB cumulative fatigue failure is proposed based on the ADLK.

The time-variant performance function $G(\mathbf{X}, t)$ of TRB is highly nonlinear and implicit. An accuracy surrogate model named Kriging is used to approximate the limit state function, $G(\mathbf{X}, t)$, to improve the efficiency of dynamic reliability assessment. The Kriging model can be

expressed as [73,74]:

$$g_k(\mathbf{X}) = \sum_{i=1}^p f_i(\mathbf{X})\beta_i + z(\mathbf{X}) \quad (17)$$

where $\mathbf{f}(\mathbf{X}) = \{f_1(\mathbf{X}), f_2(\mathbf{X}) \dots f_p(\mathbf{X})\}^T$ is the basic function of the random vector, $\boldsymbol{\beta} = \{\beta_1, \beta_2 \dots \beta_p\}^T$ is the coefficient vector of the regression model, p is the number of base functions, and $z(\mathbf{X})$ is a random Gaussian process with the mean value of zero and a variance of σ^2 . Furthermore, the covariance matrix of $z(\mathbf{X})$ can be obtained by:

$$\text{Cov}[z(\mathbf{X}^{(i)}), z(\mathbf{X}^{(j)})] = \sigma^2 [R(\mathbf{X}^{(i)}, \mathbf{X}^{(j)}, \boldsymbol{\theta})] \quad (18)$$

where $R(\mathbf{X}^{(i)}, \mathbf{X}^{(j)})$ is the auto-correlation function of the anisotropic Gaussian model, which is calculated by:

$$R[\mathbf{X}^{(i)}, \mathbf{X}^{(j)}, \boldsymbol{\theta}] = \exp\left(-\sum_{k=1}^m \theta_k |X_k^{(i)} - X_k^{(j)}|^2\right) \quad (19)$$

where $\boldsymbol{\theta} = \{\theta_1, \theta_2, \theta_3 \dots \theta_k\}^T$ is the hyperparameters vector of the auto-correlation function that can be investigated based on the maximum likelihood estimation as follows [75]:

$$\max F(\boldsymbol{\theta}) = -\frac{m \ln(\hat{\sigma}^2) + \ln(\mathbf{R})}{2}, \theta_k \geq 0 \quad (20)$$

where:

$$\begin{cases} \hat{\sigma}^2 = (\mathbf{g} - \mathbf{F}_k \hat{\boldsymbol{\beta}})^T \mathbf{R}^{-1} (\mathbf{g} - \mathbf{F}_k \hat{\boldsymbol{\beta}}) / m \\ \hat{\boldsymbol{\beta}} = (\mathbf{F}_k^T \mathbf{R}^{-1} \mathbf{F}_k)^{-1} \mathbf{F}_k^T \mathbf{R}^{-1} \mathbf{g} \end{cases} \quad (21)$$

$\hat{\boldsymbol{\beta}}$ is an estimate of $\boldsymbol{\beta}$, $\mathbf{F}_k$ is an $m \times p$ matrix composed of regression models at m sample points, and $\mathbf{g}$ is the response vector of training sample points. Therefore, the response prediction of an unknown vector $\mathbf{X}$ based on the Kriging model can be expressed as:

$$g_k(\mathbf{X}) = \mathbf{f}^T(\mathbf{X}) \hat{\boldsymbol{\beta}} + \mathbf{r}^T(\mathbf{X}) \mathbf{R}^{-1} (\mathbf{g} - \mathbf{F} \hat{\boldsymbol{\beta}}) \quad (22)$$

$$\begin{aligned} E[I(t)] &= E[\max\{G_k^{\min}(\mathbf{X}^*) - G_k(\mathbf{X}^*, t), 0\}] \\ &= (G_k^{\min}(\mathbf{X}^*) - \mu_{Gk}(\mathbf{X}^*, t)) \Phi\left(\frac{G_k^{\min}(\mathbf{X}^*) - \mu_{Gk}(\mathbf{X}^*, t)}{\sigma_{Gk}(\mathbf{X}^*, t)}\right) + \sigma(\mathbf{X}^*, t) \varphi\left(\frac{G_k^{\min}(\mathbf{X}^*) - \mu_{Gk}(\mathbf{X}^*, t)}{\sigma_{Gk}(\mathbf{X}^*, t)}\right) \end{aligned} \quad (27)$$

where $\mathbf{r}(\mathbf{X})$ is the correlation function vector between the responses of the training samples and the predicting samples. Hence, the following expression can be written:

$$\mathbf{r}^T(\mathbf{X}) = \{R(\mathbf{X}, \mathbf{X}^{(1)}), R(\mathbf{X}, \mathbf{X}^{(2)}), R(\mathbf{X}, \mathbf{X}^{(3)}) \dots R(\mathbf{X}, \mathbf{X}^{(m)})\} \quad (23)$$

Moreover, the mean and variance values of the Kriging model are derived as:

$$\begin{aligned} \mu_{gk}(\mathbf{X}) &= \mathbf{f}^T(\mathbf{X}) \hat{\boldsymbol{\beta}} + \mathbf{r}^T(\mathbf{X}) \mathbf{R}^{-1} (\mathbf{g} - \mathbf{F} \hat{\boldsymbol{\beta}}) \\ \sigma_{gk}^2(\mathbf{X}) &= \hat{\sigma}^2 \left\{ 1 - \mathbf{r}^T(\mathbf{X}) \mathbf{R}^{-1} \mathbf{r}(\mathbf{X}) + [\mathbf{F}^T \mathbf{R}^{-1} \mathbf{r}(\mathbf{X}) - \mathbf{f}(\mathbf{X})]^T (\mathbf{F}^T \mathbf{R}^{-1} \mathbf{F})^{-1} [\mathbf{F}^T \mathbf{R}^{-1} \mathbf{r}(\mathbf{X}) - \mathbf{f}(\mathbf{X})] \right\} \end{aligned} \quad (24)$$

The proposed ADLK can be divided into two adaptive learning stages: (1) The inner loop establishes the Kriging model $G_k(\mathbf{X}^*, t)$ with an even distribution of initial time samples $[t_1, t_2 \dots t_s]$ and the correlation response of $G(\mathbf{X}^*, t)$. In addition, the minimum response of $G_k(\mathbf{X}^*, t)$ is extracted as the output $G_e(\mathbf{X}^*)$ based on the optimization strategy. (2) The outer loop establishes the Kriging model $G_{ek}(\mathbf{X})$ that defines the relationship between the characteristic parameter vector $\mathbf{X}$ and extreme response $G_e(\mathbf{X})$. Dynamic reliability of wind turbine TRB can be derived from the following form:

$$P_f(0, t_s) = \text{Prob}\{G_e(\mathbf{X}) \leq 0\} \quad (25)$$

where

$$G_e(\mathbf{X}) = \min\{G(\mathbf{X}, t) \mid t \in [0, t_s]\} \quad (26)$$

Detailed steps are presented as follows:

Step 1: A random sample pool $\mathbf{S}^{\mathbf{X}} = \{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3 \dots \mathbf{X}_N\}$ is generated in the probabilistic space based on the joint probability density function $f_{\mathbf{X}}(\mathbf{X})$.

Step 2: Randomly Select the N_{in} ($N_{in} < N$) samples in $\mathbf{S}^{\mathbf{X}}$ randomly as the initial sample pool $\mathbf{S}^{\mathbf{X}i} = \{\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3 \dots \mathbf{X}_j\}$.

Step 3: The inner loop is activated to adaptively establish the Kriging model $G_k(\mathbf{X}^*, t)$ ($\mathbf{X}^* \in \mathbf{S}^{\mathbf{X}i}$) at $[0, t_s]$ (Steps 3.1 - 3.6 demonstrate how to estimate the $G_k(\mathbf{X}^*, t)$).

Step 3.1: Discretize the observation region $[0, t_s]$ to N_t time point as a time sample pool $\mathbf{S}^T = \{0, t_1, t_2 \dots t_{N^*}\}$.

Step 3.2: The N_{t^*} ($N_{t^*} < N_t$) samples are randomly extracted from $\mathbf{S}^T$ as the initial training time vector $\mathbf{t}^*$. Simultaneously, the actual responses of the time-variant performance function $G(\mathbf{X}^*, \mathbf{t}^*)$ are estimated.

Step 3.3: The Kriging model of the inner loop $G_k(\mathbf{X}^*, t)$ is established based on the $\mathbf{T}^T = \{\mathbf{t}^*, G(\mathbf{X}^*, \mathbf{t}^*)\}$.

Step 3.4: Predict the time-variant performance function value in $\mathbf{S}^T$ by $G_k(\mathbf{X}^*, t)$. Furthermore, calculate the mean $\mu_{Gk}(\mathbf{X}^*, t)$ and standard deviation $\sigma_{Gk}(\mathbf{X}^*, t)$ of the $G_k(\mathbf{X}^*, t)$ according to Eq. (24).

Step 3.5: Investigate the maximum improvement time point in $\mathbf{S}^T$ via the learning function $E[I(t)]$ that can be expressed as [76]:

where

$$G_k^{\min} = \min\{G_k(\mathbf{X}^*, t) \mid t \in [0, t_s]\} \quad (28)$$

where $\Phi(\cdot)$ and $\varphi(\cdot)$ are the cumulative distribution function (CDF) and probability density function (PDF) of standard normal variables. Therefore, the new training time point t^{up} can be updated as:

$$t^{up} = \arg \max_{t \in \mathbf{S}^T} E(I(t)) \quad (29)$$

Step 3.6: The corresponding convergence criterion of the learning function $E[I(t)]$ can be expressed as:

$$\max_{t \in S^T} E[I(t)] < C_B \quad (30)$$

where

$$C_B = 0.01 |G_k^{\min}(X^*, t)| \quad (31)$$

The inner loop is terminated by satisfying the requirement of Eq. (30). Otherwise, expand T^T according to Eq. (32) and go to Step 3.3.

$$T^T = \{T^T \cup (t^{up}, G(X^*, t^{up}))\} \quad (32)$$

Step 4: Estimate the minimum time-variant performance function on random variables fixed at S^{Xi} as follows:

$$G_e(S^{Xi}) = \min_{t \in [0, t_s]} G_k(S^{Xi}, t) \quad (33)$$

Hence, initial training samples of the outer loop can be established by $T^{ei} = \{S^{Xi}, G_e(S^{Xi})\}$.

Step 5: The Kriging model $G_{ek}(X)$ of the outer loop is established based on T^{ei} .

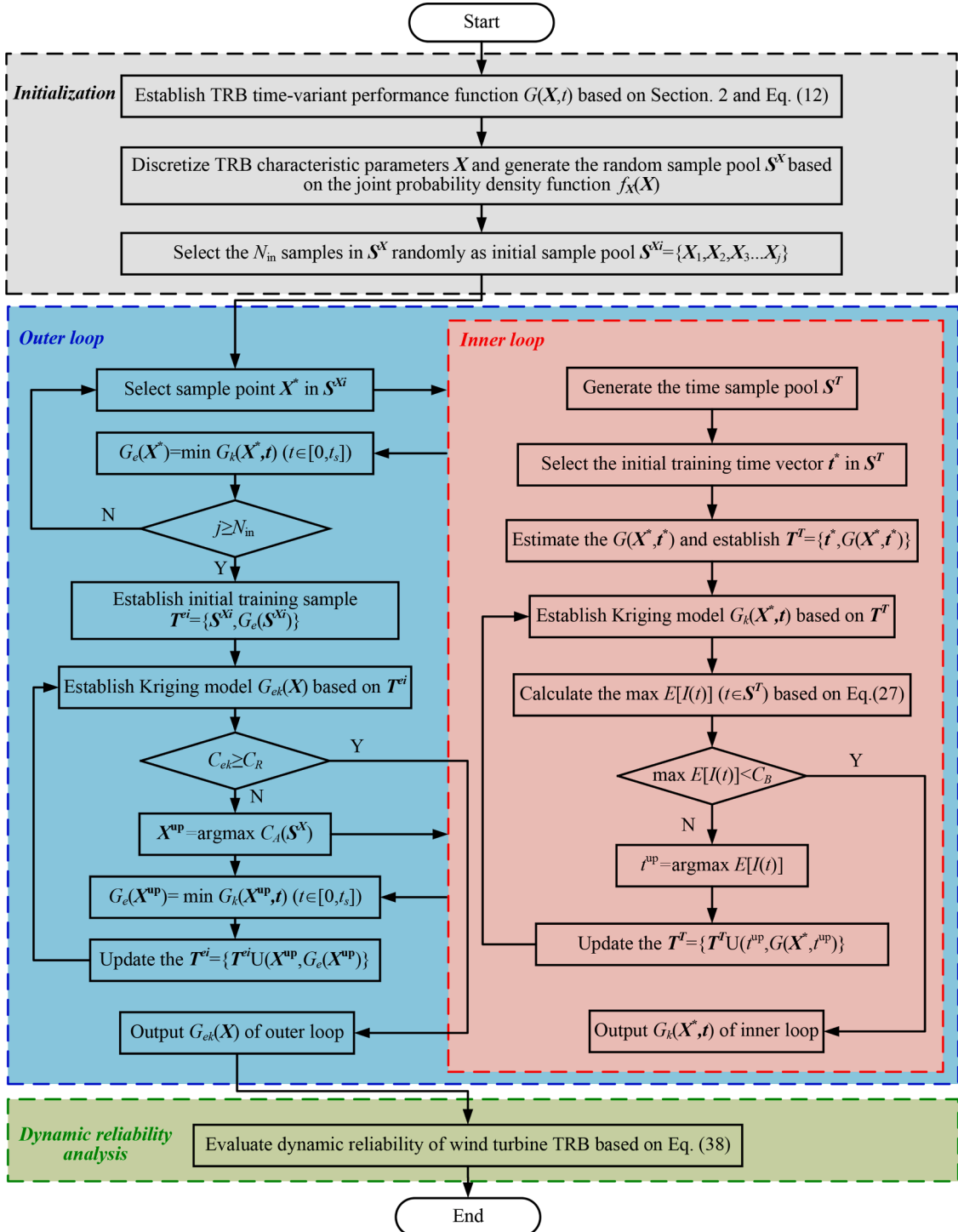


Fig. 6. Flowchart of the TRB dynamic reliability analysis based on ADLK.

Step 6: Predict the time-variant performance function value $G_{ek}(S^X)$ in the sample pool S^X by $G_{ek}(X)$. Moreover, standard deviation $\sigma_{Gek}(S^X)$ can be obtained via Eq. (24).

Step 7: Investigate the maximum improvement sample in S^X by employing the learning function $C_A(X)$:

$$C_A(S^X) = \left(1 - \Phi\left(\frac{|G_{ek}(S^X)|}{\sigma_{Gek}(S^X)}\right)\right) \times f_X(S^X) \times \sqrt{\sigma_{Gek}(S^X)} \quad (34)$$

where $f_X(X)$ is the joint probability density function of X . Therefore, the new training sample X^{up} can be updated as follows:

$$X^{up} = \underset{X \in S^X}{\operatorname{argmax}} C_A(S^X) \quad (35)$$

$$C_{ek} = \frac{1}{N} \sum_{i=1}^N \Phi\left(\frac{|G_{ek}(X_i)|}{\sigma_{Gek}(X_i)}\right) \geq C_R \quad (36)$$

where C_R is the confidence level target obtained from [77].

The outer loop is terminated by satisfying the requirement of Eq. (36). Otherwise, go to Step 3 and 4 for estimating $G_e(X^{up})$. New training samples of the outer loop can be expended as:

$$T^{ei} = \{T^{ei} \cup (X^{up}, G_e(X^{up}))\} \quad (37)$$

Step 8: Utilize the Kriging model of the outer loop $G_{ek}(X)$ to investigate the dynamic reliability of the wind turbine TRB:

$$P_f(0, t_s) = \frac{1}{N} \sum_{i=1}^N I_F(X_i) \quad (38)$$

where

$$\begin{cases} I_F(X_i) = 1, & G_{ek}(X_i) \leq 0 \\ I_F(X_i) = 0, & G_{ek}(X_i) > 0 \end{cases} \quad (39)$$

where $I_F(X_i)$ is the indicator function. The flowchart of the TRB dynamic reliability analysis is displayed in Fig. 6.

4. Numerical example

In terms of their locations, wind turbines can be categorized as onshore or offshore [3]. Due to the effect of ever-changing weather and inaccessibility, the O&M costs and failure rates of components in offshore wind turbines are higher than those in onshore ones, especially for the offshore direct drive wind turbines that are only supported by the main shaft bearing [78]. Generally, the design life of offshore wind turbines typically spans 20–25 years [79]. Therefore, as one of the core transmission parts of offshore wind turbines, the main shaft bearing (such as TRB) needs to ensure sufficient fatigue resistance during this period.

Table 2
Geometric parameters of TRB in wind turbine.

Parameter	value	Parameter	value
Outer diameter D [mm]	1580	Corner radius of the roller large-end c_1 [mm]	1.5
Bore diameter d [mm]	1270	Corner radius of the roller small-end c_2 [mm]	1.2
Roller large-end diameter D_w [mm]	75.13	Outer race contact angle α_e [°]	15.5
Roller small-end diameter d_w [mm]	71.33	Inner race contact angle α_i [°]	14
Effective contact length L_w [mm]	141	Flange angle α_f [°]	75.8
Number of rollers Z	55	Coefficient of the logarithmic crowned profile K_x	0.00045
Modulus of elasticity E [Gpa]	210	Poisson's ratio ν	0.3

In this section, different practical issues are considered to demonstrate the effectiveness and benefits of the proposed method. The main geometric parameters of the investigated object used for numerical analysis are determined based on the main shaft TRB of a 5 MW offshore direct drive wind turbine, as shown in Table 2. The distribution of structural parameters is determined according to the blueprint design error to reasonably analyze the dynamic reliability, as shown in Table 3. Simplified fatigue load cases of the wind turbine TRB are listed in Table 4. All numerical examples are tested on a personal computer with Intel i7-9700 and 16 GB of RAM.

4.1. Contact load analysis

TRB is generally subjected to external combined loads during the service period of a wind turbine. Each TRB roller generates the contact load to resist external combined loads [51,80]. This long-term cycling contact load might cause fatigue failure of the TRB. Therefore, estimating the contact load is crucial to further predict the fatigue life.

The proposed quasi-static model calculates contact loads between each roller and the outer raceway under variable fatigue load cases, as shown in Table 4. Moreover, the finite element model, Harris model [81], and Romax software are also adopted to verify the accuracy and efficiency of the proposed model. The boundary conditions of the finite element model and the Romax software are shown in Fig. 7(a), (b), and the results are depicted in Fig. 7(c)-(f).

Furthermore, the average relative error ε_{MEA} is defined according to Eq. C1 to characterize the differences between the proposed model and existing models, and the results are shown in Fig. 7(h). It is found that the ε_{MEA} of the model created by Romax software and the Harris model is less than 3%, which indicates the results obtained from the proposed model are consistent with the results obtained by the Romax software and the Harris model. However, the maximum value of ε_{MEA} obtained by the finite element model is approximately 19.1%. This deviation mainly accounts for the fact that the finite element model considers the flexibility of the TRB.

The CPU time consumption of the four used methods under the fatigue load cases A-D is summarized in Fig. 7(g). It can be seen that the CPU time is related to the complexity level of the external combined load. Moreover, the CPU time consumptions acquired by the proposed model under four load cases are 0.097 s, 0.082 s, 0.097 s, and 0.165 s. It is evident from the results that the proposed model is more efficient. Furthermore, the proposed model can be applied to the TRB with the thrust contact friction and radial clearance. However, the same cannot be stated for the Harris model, the finite element model, and the Romax software.

4.2. Fatigue life analysis

The coupling effects of nonlinear factors such as friction, clearance, and preload are essential for deeply understanding the mechanisms of TRB fatigue failure [82,83]. Therefore, the proposed quasi-static model is used to investigate the functional mechanisms of these factors on TRB fatigue life, as shown in Fig. 8.

The coupling effect of assembly preload and radial load on the fatigue life is illustrated in Fig. 8(a). For a predetermined coupling radial force, as the axial preload magnitude increases, the evolutions of TRB fatigue life can be roughly divided into four stages: a_1 , b_1 , c_1 , and d_1 , as shown in Fig. 8(d). The contact load Q_{ej} of each roller is extracted to visually compare the contact characteristics of these stages, as shown in Fig. 8(g). The results show that the load region gradually increases with the preload. Moreover, the maximum contact load first decreases and then increases. It is evident that the internal contact characteristic significantly affects the fatigue life of the TRB, as demonstrated by combining Fig. 8(d) and (g).

Fig. 8(b) depicts the coupling effect of assembly preload and radial clearance on the fatigue life. As the radial clearance increases, the

Table 3

Distribution of random variables.

Structure type	Mean	Cov	Distribution	Structure type	Mean	Cov	Distribution
Pitch diameter d_m [mm]	1425	0.001	normal	Frictional coefficient μ	0.04	0.200	normal
Roller large-end diameter D_w [mm]	75.13	0.015	normal	Radial clearance L_r [μm]	20	0.500	normal
Roller small-end diameter d_w [mm]	71.33	0.015	normal	Rotational speed n [rpm]	11	0.030	normal
Logarithmic crowned profile coefficient K_x	0.00045	0.200	normal	Effective contact length L_w [mm]	141	0.007	normal

Table 4Fatigue load cases of TRB in wind turbine with time ratio $R_{ti}=1$.

Load case	F_x [N]	F_y [N]	F_z [N]	M_y [N-mm]	M_z [N-mm]
Case A	0	1.2×10^6	0	0	2.3×10^8
Case B	0	1.2×10^6	8×10^5	1.5×10^8	2.3×10^8
Case C	0	1.2×10^6	1.2×10^6	2.3×10^8	2.3×10^8
Case D	1×10^6	1.2×10^6	8×10^5	1.5×10^8	2.3×10^8

fatigue life evolutions for a specific preload tend to decrease nonlinearly, as shown in Fig. 8(e). This phenomenon mainly accounts for the fact that the load region decreases as the radial clearance increase. In addition, the maximum internal contact load on the corresponding roller tends to

increase, as shown in Fig. 8(h). It should be noted that the proper axial preload can entirely resist the effect of radial clearance on the fatigue life of TRB, as shown in Fig. 8(b).

The coupling effect of assembly preload and friction on the fatigue life is illustrated in Fig. 8(c). For a specific assembly preload, the fatigue life evolutions trend decreases nonlinearly with an increase in the frictional coefficient, as shown in Fig. 8(f). The contact characteristic of TRB is extracted to account for this trend, as shown in Fig. 8(i). The results show that the maximum internal contact load gradually increases with the frictional coefficient. Furthermore, as the frictional coefficient increases, the load region remains constant. In addition, the emergence of the maximum fatigue life points requires a larger preload, as shown in Fig. 8(c). Notably, assembly preload cannot entirely compensate for the

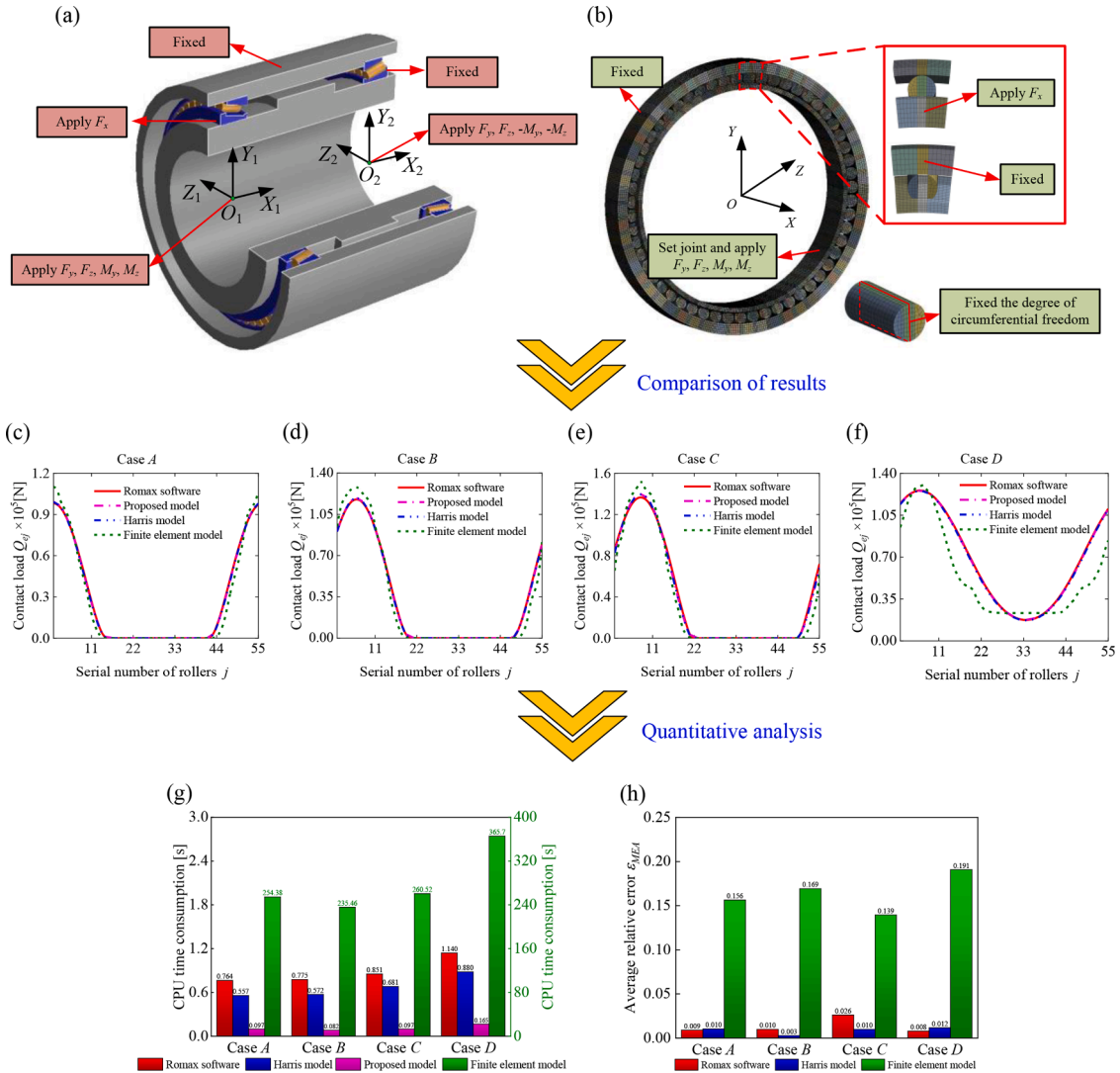


Fig. 7. Comparison between the proposed model and the existing models, which consist of finite element model, Harris model, and the Romax software: (a) boundary conditions in the Romax software, (b) boundary conditions of the finite element model. The panels in the second row show the comparison of contact load distribution under (c) case A, (d) case B, (e) case C, and (f) case D. Additional panels show the quantitative comparison of the results in (c-f). (g) the CPU time consumption of the four used models, (h) the average relative error ε_{MAE} (defined by Eq. C.1) between the proposed model and other models.

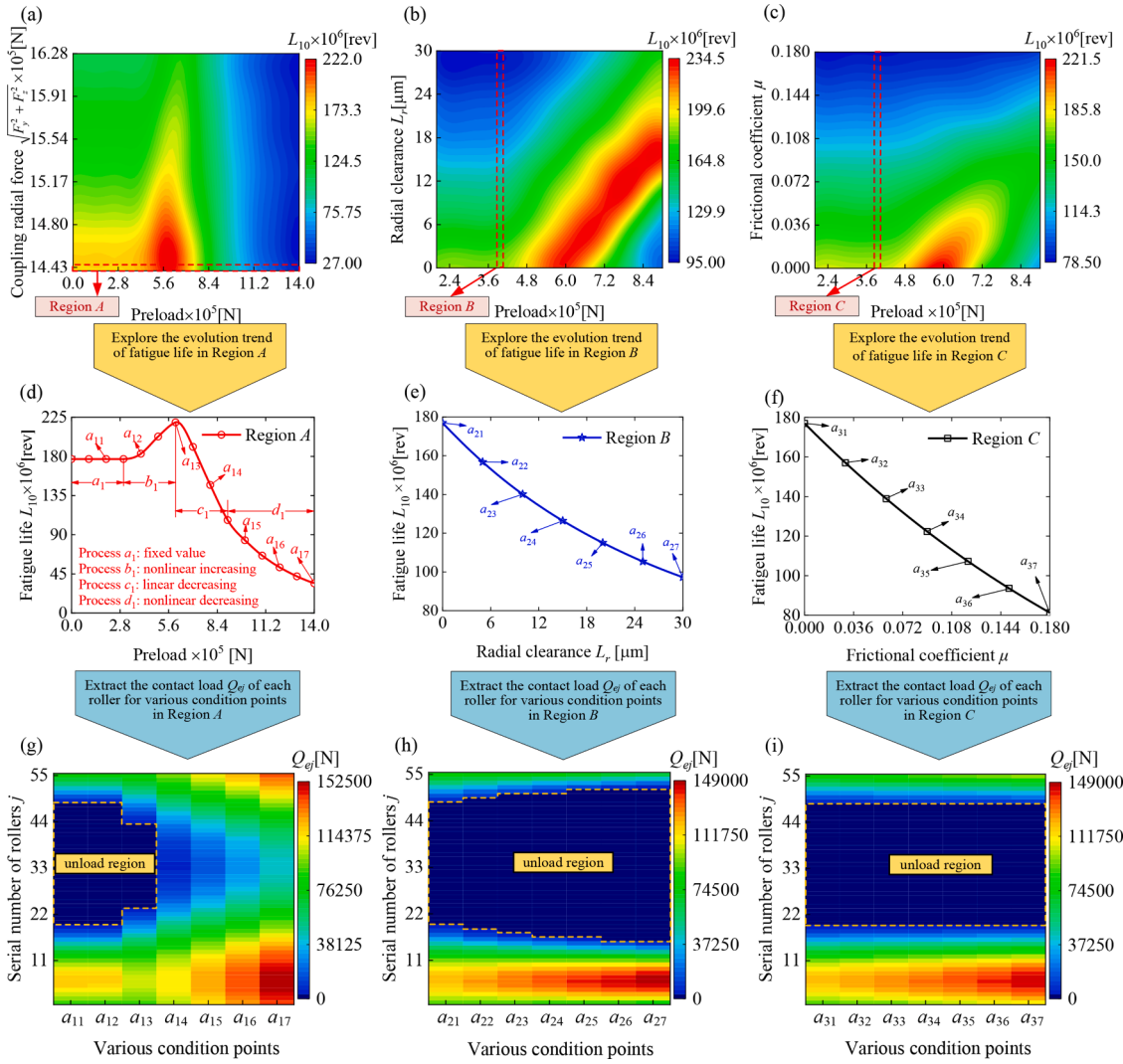


Fig. 8. The functional mechanisms of various nonlinear factors such as preload, friction, and clearance on TRB fatigue life: The coupling effect of (a) preload and radial force on TRB fatigue with $L_r = 0 \mu\text{m}$, $\mu = 0$, (b) preload and radial clearance on TRB fatigue life with $\mu = 0$ for Case B, (c) preload and friction on TRB fatigue life with $L_r = 0 \mu\text{m}$ and case B. To explore the evolutions of TRB fatigue life, the trends in (d) Region A, (e) Region B, and (f) Region C are collected. Additional panels show contact characteristic of TRB under (g) various condition points in Region A. (h) various condition points in Region B. (i) various condition points in Region C.

effect of friction on fatigue life.

4.3. Dynamic reliability analysis with constant load condition

In this section, the proposed method based on ADLK is used to estimate the dynamic fatigue failure probability of TRB and investigate the fatigue failure of TRB under the fully-discrete state and time-variant mechanism. Distribution details of random variables are shown in Table 3. A single fatigue load case B is applied to TRB, and the results are shown in Fig. 9.

The dynamic reliability of TRB is depicted in Fig. 9(b). The evolution of the fatigue failure probability throughout TRB operational life is divided into three stages: safe, centralized failure, and failure. The cumulative distributions of the time-variant performance function in diverse observation regions are extracted to account for these stages, as shown in Fig. 9(a). The results show that the cumulative distribution weight in the fatigue failure domain increases with expanding observation region, which is the primary reason for the dynamic fatigue failure probability transitions from the safe stage to the failure stage.

Variation of the dynamic fatigue failure probability is described in Fig. 9(b). The red line denotes the proposed method based on the ADLK, and the blue asterisk represents the simulation results based on the CLS.

It can be seen that the proposed method is consistent with the CLS method. In addition, the number of model evaluations required by two methods in diverse observation regions is collected, as shown in Fig. 9 (d). The results show that many model evaluation for the proposed method is concentrated on the centralized failure stage. Furthermore, the proposed method requires fewer model evaluations than CLS methods, which indicates the high efficiency of the proposed method.

It is evident that the assembly preload significantly affects the fatigue life of the TRB, as investigated in Section 4.2. The lifetime of a design project for a wind turbine is generally estimated to be 20 years [8,79, 84]. Therefore, it is necessary to investigate the effect of preload on dynamic fatigue failure probability at 20 years. Fig. 9(c) illustrates the dynamic fatigue failure probability variation with assembly preload. When the preload is increased from 0 N to 5×10^5 N, the fatigue failure probability decreases from 78.82% to 47.22% at 20 years. However, the fatigue failure probability of TRB increases as the preload exceeds 5×10^5 N. Notably, the fatigue failure probability of TRB increases sharply for the preload between 6×10^5 N and 7×10^5 N, which is undesirable and should be avoided.

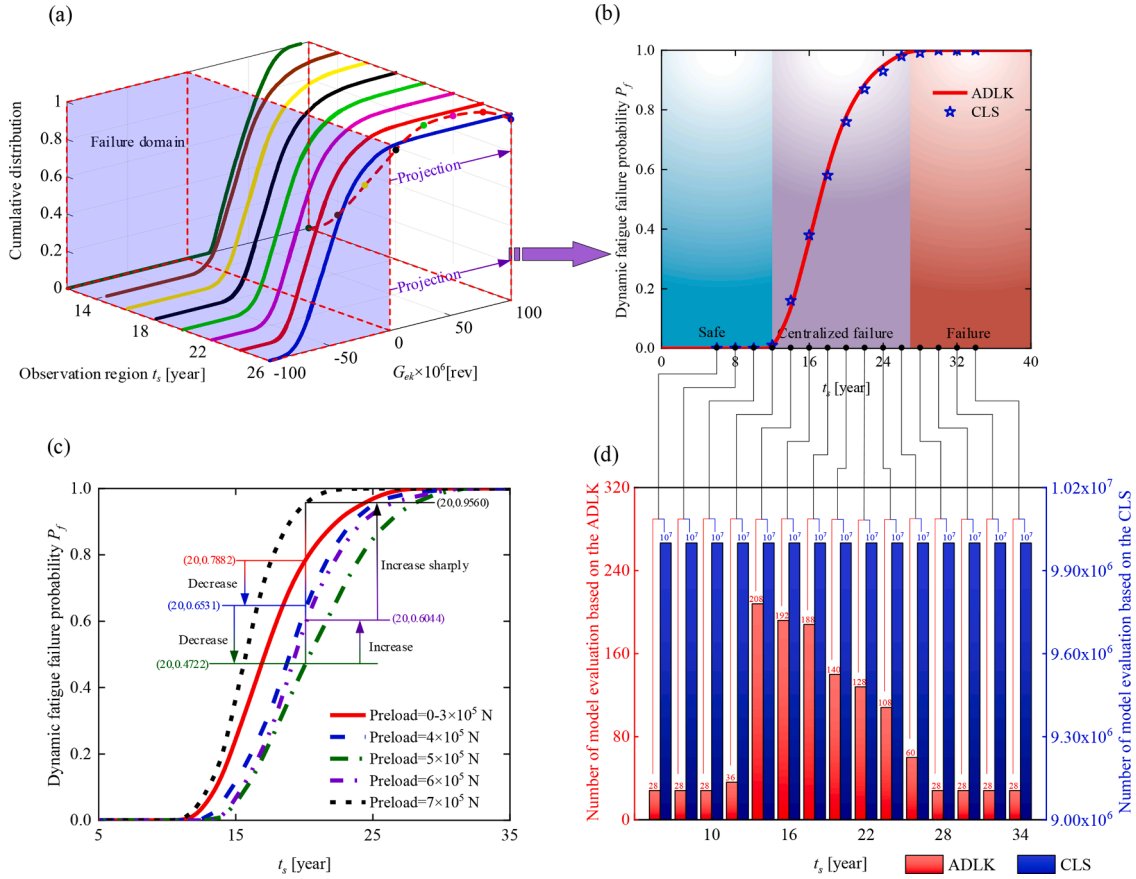


Fig. 9. An example of the TRB dynamic reliability analysis under Case B: (a) cumulative distribution of the time-variant performance function in diverse observation regions based on ADLK, (b) dynamic reliability assessment of TRB based on ADLK and CLS, (d) the number of the model evaluations based on two methods, and (c) the effect of various preloads on TRB dynamic reliability.

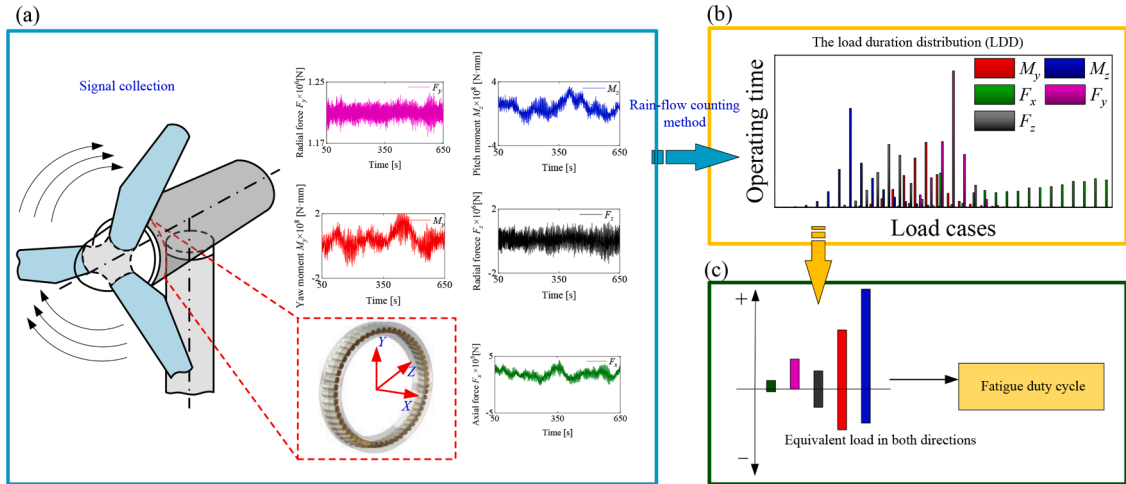


Fig. 10. Schematic diagram of fatigue duty cycle generation: (a) loading histories in the global coordinate frame XYZ of TRB, (c) generation of the load duration distribution, and (d) transformation of the equivalent loads.

4.4. Dynamic reliability with the fatigue duty cycle

The dynamic reliability of TRB under the single fatigue load case was analyzed in Section 4.3. Due to the long-term exposure to ever-changing weather stages, the TRB in wind turbines experiences a time-varying load. Therefore, carrying out the dynamic reliability analysis considering stochastic loading conditions is essential. The fatigue duty cycle procedure is illustrated in Fig. 10. Firstly, the loading histories in the

global coordinate frame XYZ of TRB are described. Subsequently, the corresponding Load Duration Distributions (LDD) are generated based on the Rain-flow counting method [85]. Finally, the fatigue duty cycle is generated in terms of equivalent load in both directions. In this section, the measured fatigue duty cycle is divided into 16 cases for the dynamic fatigue failure probability analysis and fatigue life evaluation, as shown in Table 5.

The proposed method is adopted to estimate the dynamic fatigue

Table 5
Fatigue duty cycle.

Load case i	Time ratio R_{ti}	F_x [N]	F_y [N]	F_z [N]	M_x [N·mm]	M_y [N·mm]
1	1.04×10^{-5}	-7.09×10^3	1.19×10^6	7.98×10^5	-7.19×10^7	-6.61×10^7
2	7.38×10^{-6}	-7.09×10^3	1.19×10^6	7.98×10^5	-7.19×10^7	2.28×10^8
3	3.45×10^{-6}	-7.09×10^3	1.19×10^6	7.98×10^5	1.47×10^8	-6.61×10^7
4	2.46×10^{-6}	-7.09×10^3	1.19×10^6	7.98×10^5	1.47×10^8	2.28×10^8
5	1.70×10^{-6}	-7.09×10^3	1.19×10^6	-5.32×10^5	-7.19×10^7	-6.61×10^7
6	7.19×10^{-6}	-7.09×10^3	1.19×10^6	-5.32×10^5	-7.19×10^7	2.28×10^8
7	5.11×10^{-6}	-7.09×10^3	1.19×10^6	-5.32×10^5	1.47×10^8	-6.61×10^7
8	2.39×10^{-6}	-7.09×10^3	1.19×10^6	-5.32×10^5	1.47×10^8	2.28×10^8
9	1.84×10^{-1}	3.92×10^5	1.19×10^6	7.98×10^5	-7.19×10^7	-6.61×10^7
10	8.62×10^{-2}	3.92×10^5	1.19×10^6	7.98×10^5	-7.19×10^7	2.28×10^8
11	6.13×10^{-2}	3.92×10^5	1.19×10^6	7.98×10^5	1.47×10^8	-6.61×10^7
12	2.59×10^{-1}	3.92×10^5	1.19×10^6	7.98×10^5	1.47×10^8	2.28×10^8
13	1.79×10^{-1}	3.92×10^5	1.19×10^6	-5.32×10^5	-7.19×10^7	-6.61×10^7
14	1.28×10^{-1}	3.92×10^5	1.19×10^6	-5.32×10^5	-7.19×10^7	2.28×10^8
15	5.97×10^{-2}	3.92×10^5	1.19×10^6	-5.32×10^5	1.47×10^8	-6.61×10^7
16	4.25×10^{-2}	3.92×10^5	1.19×10^6	-5.32×10^5	1.47×10^8	2.28×10^8

failure probability of TRB for a fatigue duty cycle. Furthermore, the fatigue life with the fatigue duty cycle is also investigated to characterize the distinction between the fatigue life evaluation and the dynamic reliability analysis, the results are shown in Fig. 11.

The complete process of fatigue life evaluation under the fatigue duty cycle is illustrated in Fig. 11(a). The blue columnar denotes the fatigue life with a constant load case in the fatigue duty cycle, whereas the green columnar presents the time ratio of the corresponding load case. It can be seen that the fatigue life of TRB L_{10ri} is approximately 102.48×10^6 cycles under the constant load case 12 with a maximum time ratio. However, the actual fatigue life L_d defined by Eq. (10) approaches 117.422×10^6 cycles, as shown in the black line in Fig. 11(a). This phenomenon mainly accounts for the contribution of other light-load cases. Moreover, the actual fatigue life L_d satisfies the requirement of the design target, which is defined by the number of TRB revolutions at 20 years with 11 rpm, as shown by the red line in Fig. 11(a). The dynamic fatigue failure probability analysis with the fatigue duty cycle is described in Fig. 11(b). The results show that the actual fatigue failure probability is approximately 46.64% at 20 years. Therefore, the dynamic reliability assessment of TRB proposed in this paper is more comprehensive than the fatigue life evaluation.

It should be noted that the fatigue failure probability is approximately 64.36% for the constant load case 12, as shown in Fig. 11(b). The yellow area denotes the dynamic fatigue failure probability error for a single load case 12 and fatigue duty cycle. If the duty cycle is oversimplified to the constant load case with the maximum time ratio, the fatigue life evaluation and the dynamic reliability analysis will be more conservative, which may negatively impact the maintenance strategy for the wind turbine.

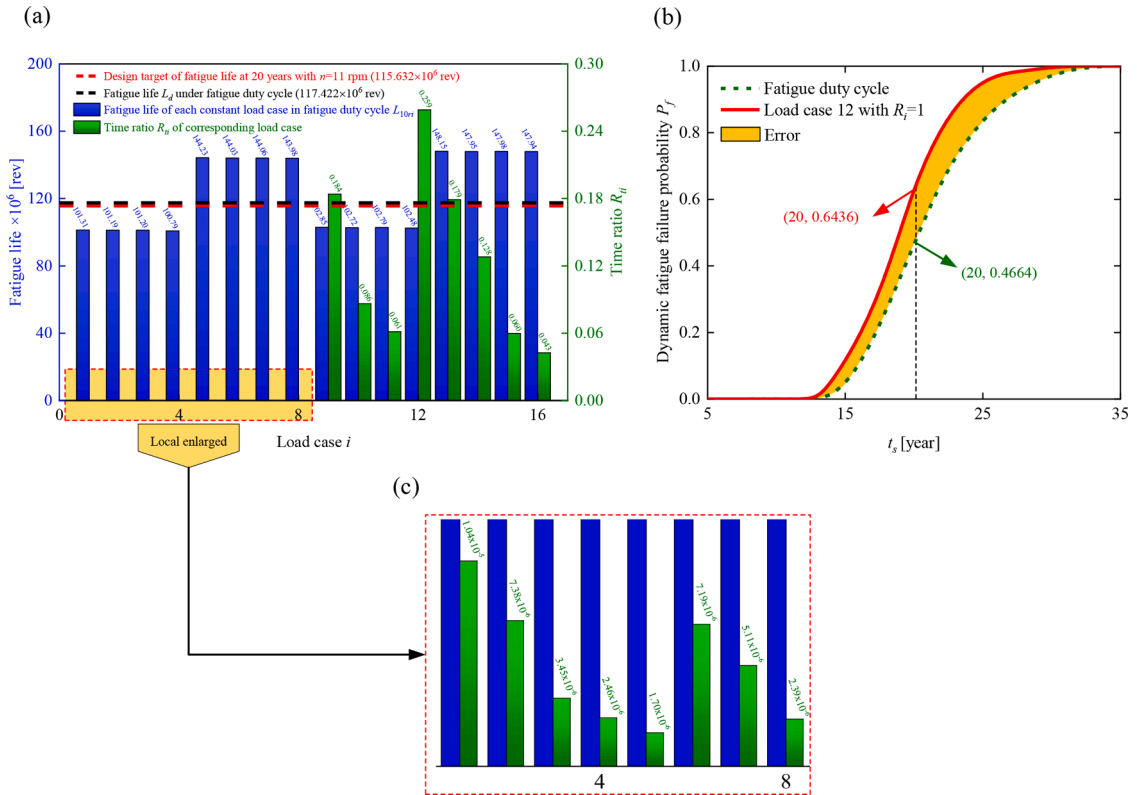


Fig. 11. Comparison between the fatigue life evaluation and dynamic failure probability analysis under the fatigue duty cycle: (a) fatigue life evaluation of each load case in the fatigue duty cycle. The fatigue life evaluation is in the blue columnar chart with $L_r = 20 \mu\text{m}$ and $\mu = 0.04$, whereas the corresponding time ratio is in the green columnar chart. The local enlarged panels of the time ratio for load cases 1 to 8 are shown in (c). (b) The comparison of dynamic failure probability analysis under load case 12 and complete fatigue duty cycle.

5. Conclusions

In this paper, high-efficiency dynamic reliability analysis and quasi-static modeling for wind turbine TRB are carried out. Dynamic reliability analysis provides valuable insights and tools for designing the main shaft bearing in wind turbines. Moreover, the proposed modeling method for TRB based on the meta-heuristic algorithm overcomes the inconvenience of the empirical initial value selection and the low efficiency of the conventional gradient-based approaches. The contributions are detailed below.

This paper presents a new TRB quasi-static model subjected to preload, bearing clearance, and friction. The derived highly nonlinear equations are solved from the perspective of meta-heuristic optimization. The accuracy of the proposed model is verified against the Harris model, finite element model, and Romax software. The CPU time consumption of the proposed model under four load cases reach up to 0.097 s, 0.082 s, 0.097 s, and 0.165 s, which indicates the proposed model has significant advantages in efficiency. In addition, the functional mechanisms of the assembly preload, bearing clearance, and friction on fatigue life of TRB are revealed based on the proposed model. An increase in the preload can improve fatigue life and eliminate the assembly clearance effect on fatigue life. However, the assembly preload cannot entirely resist the effect of friction on the TRB fatigue life.

Subsequently, dynamic reliability analysis of TRBs in wind turbines is established and subjected to the fully-discrete state and time-variant mechanisms. The accuracy of the proposed method has been verified by the CLS method. The maximum number of the model evaluations required by the proposed method is obtained as 208, which is much lower than the CLS method. Therefore, the proposed method is beneficial for reducing the time cost of the design cycle. Moreover, the effect of assembly preload on dynamic reliability is assessed. A proper preload can improve the dynamic reliability of wind turbine bearing. However, if the preload exceeded a certain level, the dynamic reliability of a wind turbine bearing will decrease sharply. Therefore, wind turbine developers should formulate the recommended assembly preload during the design and assembly phases.

Eventually, the proposed dynamic reliability analysis for wind turbine TRB is applied to the measured fatigue duty cycle. The fatigue life

evaluation under the fatigue duty cycle is also introduced for comparison. The results show that the fatigue life of TRB under the fatigue duty cycle satisfies the design target of 20 years. Nevertheless, the actual fatigue failure probability is approximately 46.64% at 20 years. Therefore, the dynamic reliability analysis of the TRB proposed in this paper is more comprehensive than the fatigue life evaluation. Furthermore, it should be noted that if the duty cycle is oversimplified, the fatigue life evaluation and the dynamic reliability analysis will be more conservative, which is undesirable and should be avoided.

CRedit authorship contribution statement

Zhiyuan Jiang: Methodology, Formal analysis, Investigation, Validation, Writing – original draft. **Xianzhen Huang:** Conceptualization, Methodology, Formal analysis, Investigation, Validation, Writing – review & editing. **Huizhen Liu:** Investigation, Writing – review & editing. **Zhiqun Zheng:** Investigation, Writing – review & editing. **Shangjie Li:** Writing – review & editing. **Shanshan Du:** Project administration, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This research is financially supported by the National Natural Science Foundation of China [grant No. 51975110], Liaoning Revitalization Talents Program [XLYC1907171] and Fundamental Research Funds for the Central Universities [N2003005, N2103005].

Appendix A. Conversion of the coordinate systems

Displacement vectors in the coordinate system $U_{xj}U_{yj}\theta$ can be expressed as:

$$\begin{cases} \delta_{U_{yj}} = \delta_{rj} \cos(\varphi_j - \varphi_{jr}) - \varphi_y x_p \sin(\varphi_j) - \varphi_z x_p \cos(\varphi_j) - \frac{L_r}{2} \\ \delta_{U_{xj}} = \delta_x + \varphi_y r_p \sin(\varphi_j) + \varphi_z r_p \cos(\varphi_j) - L_a \\ \varphi_\theta = \varphi_y \sin(\varphi_j) + \varphi_z \cos(\varphi_j) \end{cases} \quad (A.1)$$

where

$$\varphi_{jr} = \arctan(F_z / F_y) \quad (A.2)$$

$$L_a = L_r / [2 \tan(\alpha_e)] \quad (A.3)$$

Displacement vectors in the coordinate system $X_{gj}Y_{gj}\psi$ can be expressed as:

$$\begin{cases} \delta_{Y_{gj}} = \delta_{U_{yj}} \cos(\alpha_m) + \delta_{U_{xj}} \sin(\alpha_m) \\ \delta_{X_{gj}} = -\delta_{U_{yj}} \sin(\alpha_m) + \delta_{U_{xj}} \cos(\alpha_m) \\ \varphi_\psi = \varphi_\theta \end{cases} \quad (A.4)$$

The length from the point r_e to O_{enj} can be expressed as:

$$l_{eenj} = l_{oie} + l_{oiy} + l_{yenj} \quad (A.5)$$

where

$$\begin{cases} l_{oie} = d_e \cos(\varepsilon) / 2 \\ l_{yenj} = [r_p - x_p / \tan(\alpha_e)] \cos(\alpha_e) \\ l_{oij} = x_p / \sin(\alpha_e) \end{cases} \quad (\text{A.6})$$

where l_{oie} is the length from the point O_i to r_e , l_{yenj} is the length from the point O_{enj} to r_y , and l_{oij} is the length from the point O_i to r_y , as shown in Fig. 5 (c).

Appendix B. Balance equations of the TRB

The force balance equations of the TRB can be expressed as:

$$\mathbf{E}_q = \begin{bmatrix} F_x - \sum_{j=1}^Z [Q_{ej} \sin(\alpha_e) + Q_{\mu ej} \cos(\alpha_e)] \\ \sqrt{F_y^2 + F_z^2} - \sum_{j=1}^Z \{ [Q_{ej} \cos(\alpha_e) - Q_{\mu ej} \sin(\alpha_e)] \cos(\varphi_j - \varphi_{rj}) \} \\ M_y - \sum_{j=1}^Z [(M_{ej} + M_{\mu ej}) \sin(\varphi_j)] \\ M_z - \sum_{j=1}^Z [(M_{ej} + M_{\mu ej}) \cos(\varphi_j)] \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \quad (\text{B.1})$$

where Z is the number of rollers; δ_{x_0} , δ_{rj} , φ_y , and φ_z are unknown variables.

The equations of equilibrium for rollers and raceway can be expressed as:

$$\begin{cases} Q_{fij} = \frac{Q_{ej} [(1 - \mu^2) \sin(\alpha_e - \alpha_i)] - Q_g \cos(\varphi_j) [\sin(\alpha_i) + \mu \cos(\alpha_i)]}{\sin(\alpha_i + \alpha_f) + \mu \cos(\alpha_i + \alpha_f)} \\ Q_{ij} = \frac{Q_{ej} [\sin(\alpha_e + \alpha_f) + \mu \cos(\alpha_e + \alpha_f)] + Q_g \cos(\varphi_j) \sin(\alpha_f)}{\sin(\alpha_i + \alpha_f) + \mu \cos(\alpha_i + \alpha_f)} \end{cases} \quad (\text{B.2})$$

where Q_{ij} is the contact load between the j^{th} roller and the inner raceway, Q_{fij} is the end flange contact load of the j^{th} roller, Q_g is the gravity of the j^{th} roller, α_i is the contact angle of the inner raceway, and α_f is the flange angle.

Appendix C. Fatigue life assessment of the TRB

The average relative error ε_{MAE} can be expressed as:

$$\varepsilon_{MAE} = \frac{\left| \sum_{j=1}^Z \frac{Q_{Nj} - Q_{Cj}}{Q_{Cj}} \right|}{Z} \quad (\text{C.1})$$

where Q_{Nj} and Q_{Cj} are the contact loads in the load region obtained by the proposed model and the comparison model, Z is the number of rollers.

The basic dynamic load rating of the inner and outer raceway can be obtained according to [24]:

$$\begin{cases} Q_{ci} = b_m 552 \lambda \frac{(1 - \gamma)^{29/27}}{(1 + \gamma)^{1/4}} \left(\frac{\gamma}{\cos(\alpha_m)} \right)^{2/9} \left(\frac{D_w + d_w}{2} \right)^{29/27} L_w^{7/9} Z^{-1/4} \\ Q_{co} = b_m 552 \lambda \frac{(1 + \gamma)^{29/27}}{(1 - \gamma)^{1/4}} \left(\frac{\gamma}{\cos(\alpha_m)} \right)^{2/9} \left(\frac{D_w + d_w}{2} \right)^{29/27} L_w^{7/9} Z^{-1/4} \end{cases} \quad (\text{C.2})$$

where

$$\gamma = \frac{(D_w + d_w) \cos(\alpha_m)}{2d_m} \quad (\text{C.3})$$

where $b_m = 1.1$ is a factor in the basic rated dynamic loads. λ is the edge stress reduction coefficient whose value is 0.4 - 0.8. Furthermore, the equivalent dynamic load of a nonrotating and rotating raceway can be expressed as:

$$\begin{cases} Q_{ci} = \left(\frac{1}{Z} \sum_{j=1}^Z Q_{ij}^{4.5} \right)^{1/4.5} \\ Q_{eo} = \left(\frac{1}{Z} \sum_{j=1}^Z Q_{ej}^4 \right)^{1/4} \end{cases} \quad (C.4)$$

where Q_{ii} and Q_{ej} are contact loads of each roller obtained by solving Eq. (B.1).

References

- [1] Lopez-Urunuela FJ, Fernandez-Diaz B, Pagano F, et al. Broad review of "White Etching Crack" failure in wind turbine gearbox bearings: main factors and experimental investigations. *Int J Fatigue* 2021;145:106091.
- [2] Jin X, Chen YM, Wang L, et al. Failure prediction, monitoring and diagnosis methods for slewing bearings of large-scale wind turbine: a review. *Measurement* 2021;172:108855.
- [3] Schreiber A, Marx J, Zapp P. Comparative life cycle assessment of electricity generation by different wind turbine types. *J Clean Prod* 2019;233:561–72.
- [4] Dinh QV, Doan QV, Ngo-Duc T, et al. Off-shore wind resource in the context of global climate change over a tropical area. *Appl Energy* 2022;308:118369.
- [5] Papadopoulos P, Coit DW, Ezzat AA. Seizing opportunity: maintenance optimization in offshore wind farms considering accessibility, production, and crew dispatch. *IEEE Trans Sustain Energy* 2021;13(1):111–21.
- [6] Fischer K, Besnard F, Bertling L. Reliability-Centered maintenance for wind turbines based on statistical analysis and practical experience. *IEEE Trans Energy Convers* 2012;27(1):184–95.
- [7] Ossai CI, Boswell B, Davies IJ. A Markovian approach for modelling the effects of maintenance on downtime and failure risk of wind turbine components. *Renew Energy* 2016;96:775–83.
- [8] Kang J, Sun L, Soares CG. Fault Tree Analysis of floating off-shore wind turbines. *Renew Energy* 2019;133:1455–67.
- [9] Márquez FPG, Pérez JMP, Marugán AP, et al. Identification of critical components of wind turbines using FTA over the time. *Renew Energy* 2016;87:869–83.
- [10] Zhang J, Kang J, Sun L, et al. Risk assessment of floating offshore wind turbines based on fuzzy fault tree analysis. *Ocean Eng* 2021;239:109859.
- [11] Bhardwaj U, Teixeira AP, Soares CG. Reliability prediction of an offshore wind turbine gearbox. *Renew Energy* 2019;141:693–706.
- [12] Li H, Soares CG, Huang HZ. Reliability analysis of a floating off-shore wind turbine using Bayesian networks. *Ocean Eng* 2020;217:107827.
- [13] Sinha Y, Steel JA. A progressive study into offshore wind farm maintenance optimisation using risk based failure analysis. *Renew Sustain Energy Rev* 2015;42:735–42.
- [14] Scheu MN, Tremps L, Smolka U, et al. A systematic Failure Mode Effects and Criticality Analysis for offshore wind turbine systems towards integrated condition based maintenance strategies. *Ocean Eng* 2019;176:118–33.
- [15] Li H, Diaz H, Soares CG. developed failure mode and effect analysis for floating offshore wind turbine support structures. *Renew Energy* 2021;164:133–45.
- [16] Leimeister M, Kolios A. A review of reliability-based methods for risk analysis and their application in the offshore wind industry. *Renew Sustain Energy Rev* 2018;91:1065–76.
- [17] Kang J, Sun L, Sun H, et al. Risk assessment of floating offshore wind turbine based on correlation-FMEA. *Ocean Eng* 2017;129:382–8.
- [18] Zhao Q, Yuan Y, Sun W, et al. Reliability analysis of wind turbine blades based on non-Gaussian wind load impact competition failure model. *Measurement* 2020;164:107950.
- [19] Jiang Z, Hu W, Dong W, et al. Structural reliability analysis of wind turbines: a review. *Energies* 2017;10(12):2099.
- [20] Kim DH, Lee SG. Reliability analysis of offshore wind turbine support structures under extreme ocean environmental loads. *Renew Energy* 2015;79:161–6.
- [21] Wandji WN, Natarajan A, Dimitrov N. Development and design of a semi-floater substructure for multi-megawatt wind turbines at 50+ m water depths. *Ocean Eng* 2016;125:226–37.
- [22] Yang H, Zhu Y, Lu Q, et al. Dynamic reliability based design optimization of the tripod sub-structure of offshore wind turbines. *Renew Energy* 2015;78:16–25.
- [23] Velarde J, Kramhøft C, Sørensen JD, et al. Fatigue reliability of large monopiles for offshore wind turbines. *Int J Fatigue* 2020;134:105487.
- [24] Fu B, Zhao J, Li B, et al. Fatigue reliability analysis of wind turbine tower under random wind load. *Struct Saf* 2020;87:101982.
- [25] Lee YS, Choi BL, Lee JH, et al. Reliability-based design optimization of monopile transition piece for offshore wind turbine system. *Renew Energy* 2014;71:729–41.
- [26] Thapa M, Missoum S. Uncertainty quantification and global sensitivity analysis of composite wind turbine blades. *Reliab Eng Syst Saf* 2022;222:108354.
- [27] Toft HS, Branner K, Berring P, et al. Defect distribution and reliability assessment of wind turbine blades. *Eng Struct* 2011;33(1):171–80.
- [28] Liu L, Bian H, Du Z, et al. Reliability analysis of blade of the offshore wind turbine supported by the floating foundation. *Compos Struct* 2019;211:287–300.
- [29] Su HM, Kam TY. Reliability analysis of composite wind turbine blades considering material degradation of blades. *Compos Struct* 2020;234:111663.
- [30] Liu G, Liu H, Zhu C, et al. Design optimization of a wind turbine gear transmission based on fatigue reliability sensitivity. *Front Mech Eng* 2021;16(1):61–79.
- [31] Nejad AR, Gao Z, Moan T. On long-term fatigue damage and reliability analysis of gears under wind loads in offshore wind turbine drivetrains. *Int J Fatigue* 2014;61:116–28.
- [32] Yan Y. Load characteristic analysis and fatigue reliability prediction of wind turbine gear transmission system. *Int J Fatigue* 2020;130:105259.
- [33] Dong W, Nejad AR, Moan T, et al. Structural reliability analysis of contact fatigue design of gears in wind turbine drivetrains. *J Loss Prev Process Ind* 2020;65:104115.
- [34] He P, Hong R, Wang H, et al. Fatigue life analysis of slewing bearings in wind turbines. *Int J Fatigue* 2018;111:233–42.
- [35] He P, Hong R, Wang H, et al. Calculation analysis of yaw bearings with a hardened raceway. *Int J Mech Sci* 2018;144:540–52.
- [36] Wang Y, Cao J. Determination of the precise static load-carrying capacity of pitch bearings based on static models considering clearance. *Int J Mech Sci* 2015;100:209–15.
- [37] Schwack F, Bader N, Leckner J, et al. A study of grease lubricants under wind turbine pitch bearing conditions. *Wear* 2020;454:203335.
- [38] Xu J, Benson S, Wetenhall B. Comparative analysis of fatigue life of a wind turbine yaw bearing with different support foundations. *Ocean Eng* 2021;235:109293.
- [39] Yucsan YA, Viana FAC. A hybrid physics-informed neural network for main bearing fatigue prognosis under grease quality variation. *Mech Syst Signal Process* 2022;171:108875.
- [40] Liu Z, Zhang L, Carrasco J. Vibration analysis for large-scale wind turbine blade bearing fault detection with an empirical wavelet thresholding method. *Renew Energy* 2020;146:99–110.
- [41] Pan Y, Hong R, Chen J, et al. Incipient fault detection of wind turbine large-size slewing bearing based on circular domain. *Measurement* 2019;137:130–42.
- [42] Guo Y, Zhao Z, Sun R, et al. Data-driven multiscale sparse representation for bearing fault diagnosis in wind turbine. *Wind Energy* 2019;22(4):587–604.
- [43] de Azevedo HDM, Araújo AM, Bouchonneau N. A review of wind turbine bearing condition monitoring: state of the art and challenges. *Renew Sustain Energy Rev* 2016;56:368–79.
- [44] Zheng JL, Ji JC, Yin S, Tong VC. Internal loads and contact pressure distributions on the main shaft bearing in a modern gearless wind turbine. *Tribol Int* 2020;141:105960.
- [45] ISO 281. Rolling Bearings - Dynamic Load Ratings and Rating Life. 2007.
- [46] ISO/TS 16281. Rolling Bearings - Methods for Calculating the Modified Reference Rating Life for Universally Loaded Bearings. 2008.
- [47] Cretu S, Bercea I, Mitu N. A dynamic analysis of tapered roller bearing under fully flooded conditions Part 1: theoretical formulation. *Wear* 1995;188(1–2):1–10.
- [48] Han Q, Li X, Chu F. Skidding behavior of cylindrical roller bearings under time-variable load conditions. *Int J Mech Sci* 2018;135:203–14.
- [49] Zhang C, Gu L, Mao YZ, et al. Modeling the frictional torque of a dry-lubricated tapered roller bearing considering the roller skewing. *Friction* 2019;7(6):551–63.
- [50] Yang LH, Xu TF, Xu HL, et al. Mechanical behavior of double-row tapered roller bearing under combined external loads and angular misalignment. *Int J Mech Sci* 2018;142:561–74.
- [51] Tong VC, Hong SW. The effect of angular misalignment on the running torques of tapered roller bearings. *Tribol Int* 2016;95:76–85.
- [52] Cretu S, Mitu N, Bercea I. A dynamic analysis of tapered roller bearings under fully flooded conditions part 2: results. *Wear* 1995;188(1–2):11–8.
- [53] Tong VC, Hong SW. The effect of angular misalignment on the stiffness characteristics of tapered roller bearings. *Proc Inst Mech Eng Part C* 2017;231(4):712–27.
- [54] Shevitz D, Paden B. Lyapunov stability theory of nonsmooth systems. *IEEE Trans Autom Control* 1994;39(9):1910–4.
- [55] Heidari AA, Mirjalili S, Faris H, et al. Harris hawks optimization: algorithm and applications. *Fut Gener Comput Syst* 2019;97:849–72.
- [56] Guo Y, Sheng SW, Phillips C, et al. A methodology for reliability assessment and prognosis of bearing axial cracking in wind turbine gearboxes. *Renew Sustain Energy Rev* 2020;127:109888.
- [57] Sheng SS. Report on Wind Turbine Subsystem Reliability-A Survey of Various Databases. CO: Golden; 2013. Natl Renew Energy Lab (NREL)/Tech Rep 2013.
- [58] Torsvik J, Nejad AR, Pedersen E. Experimental field study of floater motion effects on a main bearing in a full-scale spar floating wind turbine. *Mar Struct* 2021;79:103059.
- [59] Wang Y, Guo Y, Zhang D, et al. Analysis and mitigation of the drive train fatigue load for wind turbine with inertial control. *Int J Electr Power Energy Syst* 2022;136:107698.
- [60] Dhanola A, Garg HC. Tribological challenges and advancements in wind turbine bearings: a review. *Eng Fail Anal* 2020;118:104885.
- [61] Yucsan YA, Viana FAC. A physics-informed neural network for wind turbine main bearing fatigue. *Int J Prognost Health Manag* 2020;11(1).

- [62] Turi MB, Marks CS. Bearing Selection Techniques as Applied to Main Shaft Direct and Hybrid Drives for Wind Turbines. 2010. Timken Technical Paper.
- [63] Fatemi A, Yang L. Cumulative fatigue damage and life prediction theories: a survey of the state of the art for homogeneous materials. *Int J Fatigue* 1998;20(1):9–34.
- [64] Zhu SP, Liu Q, Lei Q, et al. Probabilistic fatigue life prediction and reliability assessment of a high pressure turbine disc considering load variations. *Int J Damage Mech* 2018;27(10):1569–88.
- [65] Dong Y, Teixeira AP, Soares CG. Application of adaptive surrogate models in time-variant fatigue reliability assessment of welded joints with surface cracks. *Reliab Eng Syst Saf* 2020;195:106730.
- [66] Qian HM, Li YF, Huang HZ. Time-variant reliability analysis for industrial robot RV reducer under multiple failure modes using Kriging model. *Reliab Eng Syst Saf* 2020;199:106936.
- [67] Li X, Chen G, Cui H, et al. Direct probability integral method for static and dynamic reliability analysis of structures with complicated performance functions. *Comput Methods Appl Mech Eng* 2021;374:113583.
- [68] Singh A, Mourelatos ZP, Li J. Design for lifecycle cost using time-dependent reliability. *J Mech Des* 2010;132(9):091008.
- [69] Wang Z, Wang P. A nested extreme response surface approach for time-dependent reliability-based design optimization. *J Mech Des* 2012;134(12).
- [70] Zhang XB, Lu ZZ, Cheng K. Reliability index function approximation based on adaptive double-loop Kriging for reliability-based design optimization. *Reliab Eng Syst Saf* 2021;216:108020.
- [71] Wang Z, Wang P. A new approach for reliability analysis with time-variant performance characteristics. *Reliab Eng Syst Saf* 2013;115:70–81.
- [72] Li J, Chen J. Solving time-variant reliability-based design optimization by PSO-t-IRS: a methodology incorporating a particle swarm optimization algorithm and an enhanced instantaneous response surface. *Reliab Eng Syst Saf* 2019;191:106580.
- [73] Kaymaz I. Application of kriging method to structural reliability problems. *Struct Saf* 2005;27:133–51.
- [74] Zhao Z, Lu ZH, Zhao YG. An efficient extreme value moment method combining adaptive Kriging model for time-variant imprecise reliability analysis. *Mech Syst Signal Process* 2022;171:108905.
- [75] Dutta S, Ghosh S, Inamdar MM. Optimisation of tensile membrane structures under uncertain wind loads using PCE and kriging based metamodels. *Struct Multidiscip Optim* 2018;57(3):1149–61.
- [76] Jones DR, Schonlau M, Welch WJ. Efficient global optimization of expensive black-box functions. *J Glob Optim* 1998;13(4):455–92.
- [77] Wang ZQ, Wang PF. A double-loop adaptive sampling approach for sensitivity-free dynamic reliability analysis. *Reliab Eng Syst Saf* 2015;142:346–56.
- [78] Van de Kaa G, Van Ek M, Kamp LM, et al. Wind turbine technology battles: gearbox versus direct drive-opening up the black box of technology characteristics. *Technol Forecast Soc Change* 2020;153:119933.
- [79] Topham E, McMillan D, Bradley S, et al. Recycling offshore wind farms at decommissioning stage. *Energy Policy* 2019;129:698–709.
- [80] Zhao LH, Li QC, Feng JZ, et al. Service life prediction method for wheel-hub-bearing under random multi-axial wheel loading. *Eng Fail Anal* 2021;122:105211.
- [81] Harris TA, Mindel MH. Rolling element bearing dynamics. *Wear* 1973;23(3):311–37.
- [82] El Laithy M, Wang L, Harvey TJ, et al. Further understanding of rolling contact fatigue in rolling element bearings-a review. *Tribol Int* 2019;140:105849.
- [83] Li B, Li P, Zhou R, et al. Contact mechanics in tribological and contact damage-related problems: a review. *Tribol Int* 2022:107534.
- [84] Bicer Y, Khalid F. Life cycle environmental impact comparison of solid oxide fuel cells fueled by natural gas, hydrogen, ammonia and methanol for combined heat and power generation. *Int J Hydrog Energy* 2020;45(5):3670–85.
- [85] Amzallag C, Gery JP, Robert JL, et al. Standardization of the rainflow counting method for fatigue analysis. *Int J Fatigue* 1994;16(4):287–93.